SOURCE CODE ANALYSIS

Breaking the Algorithm

A Growth Guide for Small X Accounts

Based on the Open-Sourced Code

Every recommendation in this guide cites specific code variables, weights, and mechanisms from X's open-sourced recommendation algorithm.

Sources: `twitter/the-algorithm` (2023) · `xai-org/x-algorithm` (2026)

150x	3	-738	4x
REPLY VALUE VS LIKE	TWEETS SCORED < 65	LIKES PER REPORT	PREMIUM BOOST

March 2028 · Generated from 4,331 lines of algorithm analysis

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Your Starting Position

YOUR ACCOUNT PROFILE

96 followers · 104 following · Account created 2023 · Free tier · Dormant — now reactivating

The algorithm sees you as a ghost. Here's the math.

TweepCred: Your Reputation Score

TweepCred is a daily-computed PageRank score (0–100) from `src/scala/com/twitter/graph/batch/job/tweepcred/`. It determines how many of your tweets the algorithm even **considers**.

YOUR ESTIMATED SCORE

25–40

out of 100

CRITICAL THRESHOLD

65

below this = only 3 tweets scored

From `ranking.thrift`:

```
antiGamingMinTweepcred = 65
maxHitsPerUser = 3
```

WHAT THIS MEANS

Below TweepCred 65, the EarlyBird search system only considers **3 of your tweets** for distribution. You could post 10 excellent tweets — 7 are invisible to the ranker. Threads are excluded entirely.

Your Ratio Is Okay (For Now)

From `Reputation.scala`, the penalty formula:

```
division_factor = exp(5 * (ratio - 0.6))
// where ratio = followings / followers
```

SCENARIO	FOLLOWING	FOLLOWERS	RATIO	PENALTY
Worst case (300/100)	300	100	3.0	÷ 163,000
You (actual)	104	96	1.08	None*
Target	60	96	0.63	None
Optimal	50	96	0.52	Bonus

*No penalty triggers because you're under the 500-following hard gate in `UserMass.scala`

GOOD NEWS

Your 1.08 ratio doesn't trigger the penalty — the `UserMass.scala` threshold requires 500+ following. Your main problem is dormancy (zero Real Graph edges), not ratio.

The Cold Start Problem

Real Graph Edges: ZERO

After dormancy, all interaction edges have decayed to zero. Edge formula uses exponential time decay. Your tweets **won't surface** in followers' feeds because Real Graph predicts zero interaction probability.

Source:
`src/scala/com/twitter/interaction_graph/`

SimClusters: NOT IN KNOWN-FOR

Only top 20M producers are in the Known-For matrix. At 96 followers, you're excluded. Your tweets have no pre-computed community embedding and can't be recommended via out-of-network discovery.

Source:
`src/scala/com/twitter/simclusters_v2/`

THE VICIOUS CYCLE

Low TweepCred → Only 3 tweets scored → Low Real Graph weights → Tweets don't surface → No engagement → No SimClusters embedding → No discovery → TweepCred stays low → Repeat

Decision: Keep Your 2023 Account

VERDICT: DO NOT CREATE A NEW ACCOUNT

- Account age factor `min(1.0, log(1 + age/15))` caps at 1.0 after 30 days — your 3-year account and a 2-month account score the same
- **But** your 96 existing followers are dormant Real Graph edges that can be **reactivated**
- New accounts start with zero edges entirely
- New accounts face stricter anti-spam heuristics
- **Old dormant account is strictly better than new**

Step 1 — Trim Your Following List

Your ratio isn't penalizing you yet, but trimming it improves TweepCred and signals authority.

104	~60	10-15
CURRENT FOLLOWING	TARGET FOLLOWING	UNFOLLOWS PER DAY

Exact actions:

1. Open your following list
2. Unfollow everyone you don't genuinely read or interact with
3. **Speed limit: 10–15 unfollows per day** — mass unfollow triggers a 3-month shadowban
4. Target: 60 following (ratio 0.63, essentially neutral)
5. This takes 3-4 days at safe pace
6. **Never follow someone unless you'll actually engage with them**

> **WARNING: DO NOT RUSH**
>
> Mass unfollow = 3-month reduced distribution from the anti-gaming system. 10-15 per day is safe. Do not do 44 in one sitting.

Step 2 — Pick Your 10–15 Reply Targets

This is the highest-leverage action in the entire algorithm.

The Weight Table

ENGAGEMENT SIGNAL	WEIGHT	VS LIKE
Reply + author engages back	**+75.0**	**150x**
Reply	**+13.5**	**27x**
Profile click + engage	+12.0	24x
Good click (stay 2+ min)	+11.0	22x
Retweet	+1.0	2x
Like (baseline)	+0.5	1x
Block / mute / “show less”	–74.0	–148x
Report	**–369.0**	**–738x**

THE MATH OF ONE GOOD REPLY EXCHANGE

Your reply: `+13.5 × P(reply)`
Reader clicks your profile: `+12.0 × P(profile_click)`
Author replies back: `+75.0 × P(reply_engaged)`
Total: 100 weight units

vs. one like on your original tweet: `+0.5 × P(like)` = **0.5 weight units**

Ratio: 200x more algorithmic value from one reply exchange

Criteria for Choosing Targets

- ✓ **In your niche** (longevity, biotech, geroscience, health)
- ✓ **1K–50K followers** — big enough for active threads, small enough to notice you
- ✓ **Posts at least 3x/week** — consistent activity
- ✓ **Actually replies to people** — check their recent replies tab
- ✗ Accounts with 500K+ followers — they won’t see your reply
- ✗ Accounts that never reply back to anyone

Write down 15 accounts. These are your daily reply targets for the next month.

Step 3 — Daily Reply Routine (Weeks 1–4)

KEY INSIGHT: REPLIES DON'T COUNT AGAINST YOUR 3-TWEET CAP

The `antiGamingMinTweepcred = 65` `maxHitsPerUser = 3` cap only applies to your original tweets in the ranking pipeline. **You can reply unlimited times.** This is your primary growth lever.

Time Commitment: 45–60 Minutes/Day

Morning routine (before posting anything original):

1. Open your 15 target accounts' recent posts
2. Find 3–5 posts where you have a genuinely useful reply
3. Write **substantive** replies — not "great post!" or emoji reactions

What a Good Reply Looks Like

BAD — Zero algorithmic value

"Great thread! 🔥"

"This 👆"

"So true!"

No reply-back. No profile click. No Real Graph edge. Wasted time.

GOOD — Generates author reply-back

"The rapamycin dosing data here is interesting but the PEARL trial used 5mg weekly vs the 1mg daily in that Nature paper — completely different pharmacokinetics. Weekly pulse dosing avoids mTORC2 inhibition which is where the immunosuppression risk comes from."

Author likely replies → +75.0 weight. Readers click profile → +12.0 each.

Building Real Graph Edges

Each reply exchange creates a Real Graph edge. The features that matter:

REALGRAPH FEATURE	HOW TO BUILD IT
`nonZeroDays`	Interact with the **same** people every single day
`ewmaDecayedCount`	Multiple interactions per day compounds the signal
`daysSinceLastInteraction`	Never let more than 1–2 days pass without interacting
`diversityMetric`	Mix replies, likes, retweets, quote tweets with same person
`commonFriends`	Follow people your targets follow

Source: `src/scala/com/twitter/interaction_graph/` — visible interactions get 5x weight vs implicit (clicks, views)

Step 4 — Original Tweets (2–3 Per Day)

2–3

TWEETS PER DAY

All 3 get scored. Post 10 and the algorithm randomly picks 3.

0

THREADS

Each thread tweet counts against your 3-tweet cap. Wait until Tweep-Cred > 65.

2x

MEDIA BOOST

EarlyBird applies 2x for images/video at candidate retrieval.

Tweet Format Rules

DO	DON'T	CODE REASON
Text + image	Bare link	Free account links = 0% median engagement
1 niche hashtag	3+ hashtags	`multipleHashtagsOrTrendsDampening` = −40%
Standalone post	Thread	3-tweet cap wastes slots on thread tweets
Strong opinion/take	Bland observation	No engagement = no SimClusters embedding
Original image/chart	No media	Missing the 2x EarlyBird media boost

Ideal Tweet Structure

[Strong opening opinion or insight]

[2–3 sentences of evidence or analysis]

[Image: chart, paper screenshot, data visualization]

#Longevity

That's it. No thread. No link. One hashtag. Image attached.

IF YOU MUST SHARE A LINK

Post a text-only tweet with your take first. Then reply **to yourself** with the link. The parent tweet gets full algorithmic treatment. The link lives in the reply where it doesn't trigger the link penalty on your main tweet.

Step 5 — Timing & The 30-Minute Window

Tweet Decay Curve

From `ThriftAgeDecayRankingParams` in `ranking.thrift`:

```
ageDecayHalflife = 360.0    // 6 hours
ageDecaySlope    = 0.003
ageDecayBase     = 0.6      // floor — tweets never go below 60%
```

TIME AFTER POSTING	SCORE	STATUS
0–30 minutes	**100%**	**CRITICAL WINDOW — engagement here determines everything**
30 minutes	97%	Still strong
6 hours	50%	Half relevancy gone
12 hours	25%	Fading fast
24 hours	6%	Effectively dead

THE FIRST 30 MINUTES

Early engagement signals are weighted most heavily. The algorithm uses early velocity to predict total engagement. Tweets with traction in the first 30 minutes receive amplified distribution. Tweets with zero engagement in 30 minutes are effectively dead.

Example Daily Schedule

9:00 AM ET	Post tweet #1 (your best content of the day)
9:00–9:30	Reply to responders on your tweet **immediately**
9:30–10:00	Reply to 2–3 of your target accounts' posts
12:00 PM ET	Post tweet #2
12:00–12:30	Same: engage with responders immediately
2:00 PM ET	Reply session on target accounts' afternoon posts

Best days: Tuesday–Thursday. Best hours: 9 AM – 3 PM ET for US/EU biotech audience.

Step 6 — Shape Your SimClusters Identity

From `src/scala/com/twitter/simclusters_v2/README.md`:

> *"The InterestedIn vector of each user who Fav-ed the tweet is added to the tweet vector."*

Your **likes** shape your SimClusters InterestedIn vector. This determines which community the algorithm puts you in.

What to Do

- ✓ **Like 10–15 tweets/day ONLY from accounts in your niche** (longevity, biotech, geroscience)
- ✓ When community members like YOUR tweets, their vectors embed your tweet into the community
- ✓ This is invisible work but directly shapes what the algorithm thinks you are
- ✗ **Do NOT like random viral content** — every like shifts your InterestedIn vector
- ✗ Don't like memes, politics, sports if your niche is biotech — dilutes your cluster signal

HOW DISCOVERY WORKS FOR SMALL ACCOUNTS

You don't need to be in the Known-For matrix (top 20M) for your tweets to be discovered:

1. You post a tweet relevant to a topic community
2. Even 2–3 favorites from users in that SimCluster embed your tweet
3. Your tweet surfaces as out-of-network candidate for other community members
4. This is **engagement-driven, not follower-driven** — your follower count is irrelevant

Step 7 — Build Dense, Not Broad

GraphJet (`src/java/com/twitter/recos/`) powers 15% of For You via SALSA random walks.

How It Works

Your Circle of Trust (5–10 accounts you engage with most)

↓ random walk ↓

Tweets they engaged with

↓ random walk ↓

Other users who engaged with those tweets

↓ and vice versa ↓

Those users discover YOU through the same path

THE STRATEGY

Pick **5–10 accounts** in your niche. Engage with them **daily** — reply, like, retweet. The goal is dense Real Graph edges that GraphJet's random walks traverse efficiently.

5 strong edges beat 50 weak ones. Don't spread thin across hundreds of accounts.

Step 8 — What to NEVER Do

ACTION	WEIGHT	LIKES TO OFFSET	RECOVERY
Get reported	**−369.0**	**738 likes**	Months
Get blocked / muted	−74.0	148 likes	140-day decay
Mass follow then unfollow	N/A	N/A	3+ months
Post bare link (free)	0 reach	N/A	Immediate
Use 3+ hashtags	−40%	N/A	Immediate
Post 4+ tweets/day	Wasted	—	—
Post threads	Wasted	—	—
Follow 500+ accounts	exp(5×(r-0.6))	Destroys TweepCred	Permanent
Reply “great post!”	Zero	—	—
Like random viral content	Dilutes SimClusters	—	Permanent drift

THE ASYMMETRY IS BRUTAL

The algorithm is architecturally biased toward punishment over reward:

- **One report** (-369.0) requires **738 likes** (+0.5 each) to offset
- **One “show less”** (-74.0) requires **148 likes** to offset
- At 96 followers, even **one** negative signal can be catastrophic

Source: `FeedbackFatigueScorer.scala` — negative signals decay over 140 days with 0.2x–1.0x multiplier

Step 9 — When to Get Premium

From `HomeGlobalParams.scala`:

```
BlueVerifiedAuthorInNetworkMultiplierParam   = 4.0
BlueVerifiedAuthorOutOfNetworkMultiplierParam = 2.0
```

Plus +4 to +16 TweepCred bonus.

CONDITION	SUBSCRIBE?	WHY
Week 1–4, < 200 followers	**NO**	4x × zero Real Graph edges = zero
You post links regularly	**YES, NOW**	Free accounts get 0% engagement on links
TweepCred estimated 50–60	**YES**	+4-16 bonus can bridge past the 65 threshold — **biggest unlock**
500+ followers, consistent engagement	**YES**	4x on a real audience compounds into 10x reach

THE 4x MULTIPLIER IS ONLY AS GOOD AS YOUR BASE

Premium multiplies your reach. But 4x × 0 = 0. Build Real Graph edges first. Subscribe when the multiplier has something to multiply.

Milestones & Timeline

MILESTONE	EXPECTED	WHAT UNLOCKS
Ratio < 1.0	Day 4–5	TweepCred penalty removed
First reply-back from target	Day 2–3	150x like value + Real Graph edge
10 active Real Graph edges	Week 2	Your tweets visible in followers' feeds
250 followers	Week 3–4	Meaningful in-network audience
Tweets getting 50+ impressions	Week 3–4	Algorithmic distribution starting
TweepCred > 65	**Month 2–4**	**THE BIG UNLOCK: unlimited tweets scored**
First out-of-network viral tweet	Month 2–3	SimClusters embedding working
1,000 followers	Month 4–6	Self-sustaining growth flywheel

The Flywheel (After TweepCred 65)

All tweets considered (no 3-tweet cap)

↓

Better tweets surface (Heavy Ranker sees full output)

↓

More engagement on better tweets

↓

Stronger SimClusters embeddings

↓

More out-of-network discovery → More followers → Better TweepCred → **Repeat**

Everything before the flywheel is grinding. Everything after it compounds.

The Complete Scoring Formula

The Heavy Ranker (MaskNet, 48M parameters) scores every candidate tweet:

```
score = SUM(weight_i × P(engagement_i))
```

```
Heavy Ranker Score =
  P(reply + author_engaged) × 75.0
 + P(reply) × 13.5
 + P(profile_click + engagement) × 12.0
 + P(good_click + 2m stay) × 11.0
 + P(bookmark) × 10.0
 + P(retweet) × 1.0
 + P(favorite) × 0.5
 + P(video_watch_50%) × 0.005
 + P(negative_reaction) × -74.0
 + P(report) × -369.0
```

Post-ranking adjustments layered on top:

ADJUSTMENT	FACTOR	SOURCE
Out-of-network scaling	0.75x	`OONTweetScalingScorer.scala`
Reply scaling	0.75x	`ScoredTweetsParam.scala`
Author diversity decay	0.5x per successive tweet	`ScoredTweetsParam.scala`
Premium in-network boost	4.0x	`HomeGlobalParams.scala`
Premium out-of-network boost	2.0x	`HomeGlobalParams.scala`
Feedback fatigue	0.2x – 1.0x	`FeedbackFatigueScorer.scala`

The Pipeline — How “For You” Works

500M daily tweets enter the system

↓

1,500 **Candidate Sourcing** — 50% in-network (EarlyBird) + 50% out-of-network (SimClusters, GraphJet)

↓

500 **Light Ranker** — logistic regression pre-filter (TweepCred, engagement, content quality)

↓

100 **Heavy Ranker** — MaskNet neural network, 48M params, 10 engagement predictions

↓

50 **Post-Ranking** — OON scaling, Premium boost, diversity, feedback fatigue

↓

50 **Your “For You” feed**

WHERE SMALL ACCOUNTS GET FILTERED OUT

Stage 1 — Candidate Sourcing: Without Real Graph edges, your tweets don’t enter the in-network pool. Without SimClusters presence, you’re excluded from out-of-network.

Stage 2 — Light Ranker: TweepCred < 65 means only 3 tweets pass. Low engagement signals mean low pre-filter scores.

Stage 3 — Heavy Ranker: Even if you reach here, zero engagement history means low predicted P(engagement) across all signals.

The reply strategy bypasses all of this — replies appear in the thread itself, not through the ranking pipeline.

Source Code References

Primary Repositories

REPO	DESCRIPTION
`twitter/the-algorithm`	2023 open-source release (Scala/Java) — full recommendation pipeline
`twitter/the-algorithm-ml`	ML models including Heavy Ranker (MaskNet) weights
`xai-org/x-algorithm`	2026 Grok-powered replacement (Rust) — Phoenix, weighted_scorer

Key Files Referenced

`Reputation.scala`	TweepCred ratio penalty, PageRank mapping
`UserMass.scala`	Initial mass: device, age, ratio factors
`WeightedPageRank.scala`	Iterative PageRank (damping=0.1)
`ranking.thrift`	antiGamingMinTweepcred=65, decay params
`HomeGlobalParams.scala`	Premium 4x/2x multipliers
`OONTweetScalingScorer.scala`	Out-of-network 0.75x scaling
`FeedbackFatigueScorer.scala`	140-day negative signal decay
`ScoredTweetsParam.scala`	All scoring parameters
`SimClusters v2 README`	145K communities, Known-For matrix
`Interaction Graph README`	Real Graph edge features
`weighted_scorer.rs`	2026 Grok 19-metric scoring

Generated from 4,331 lines of algorithm analysis across 6 documents.
Every recommendation cites specific source code variables and weights.

Full analysis: `docs/x-algorithm-analysis/`